

SOLUTIONS

BILKENT UNIVERSITY
ELECTRICAL AND ELECTRONICS ENGINEERING

MATH 241
ENGINEERING MATHEMATICS I

MIDTERM EXAM

26-11-2013

Instructions:

- There are five questions in the exam. They have equal weights. Solve all.
 - Duration of the exam is 120 minutes.
 - The exam is closed book, closed notes.
 - No calculators are allowed.
 - Please read each question carefully and explain your answers clearly in order to get credit.
 - Do not detach pages.
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Surname:.....

Name:.....

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Section:.....

Signature:.....

Question	Grade
1	
2	
3	
4	
5	
Total:	

Q1. (20 points) Consider the following linear system of equations:

$$x_1 + 4x_2 - x_3 = 1$$

$$x_1 + 6x_2 - 3x_3 = 5$$

$$2x_2 + ax_3 = b$$

(a) Determine all values of the constants a and b for which the given system has

- (i) no solution;
- (ii) a unique solution;
- (iii) an infinite number of solutions.

(b) Let the solution set be given as below, what are a, b, c ?

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} c \\ 2 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \quad \alpha \in \mathbb{R}$$

$$(a) \begin{bmatrix} 1 & 4 & -1 & 1 \\ 1 & 6 & -3 & 5 \\ 0 & 2 & a & b \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 4 & -1 & 1 \\ 0 & 2 & -2 & 4 \\ 0 & 2 & a & b \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 4 & -1 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & a+2 & b-4 \end{bmatrix} \quad (4 \text{pts})$$

(i) if $a = -2$ and $b \neq 4$ there is no solution (2 points)

(ii) if $a \neq -2$ we have a unique solution (2 points)

(iii) if $a = -2$ and $b = 4$ we have infinitely many solutions (2 points)

(b) Since we have infinitely many solutions $a = -2$ and $b = 4$

4 points

$$\begin{bmatrix} 1 & 4 & -1 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & 3 & -7 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 = -7 - 3\alpha$$

$$x_2 = 2 + \alpha$$

$$x_3 = \alpha$$

$$= \begin{bmatrix} c \\ 2 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \boxed{c = -7}$$

6 points

Q2. (20 points)

(a) Consider the 4×4 matrix given below

$$A = \begin{bmatrix} -1 & 1 & -1 & 8 \\ -1 & -1 & 9 & 7 \\ 0 & 0 & 2 & -3 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

Compute $B = A^4 + 2A^3 - 2A^2 - 9A - 6I$

(b) Consider the 3×3 matrix given below

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & \frac{\pi}{2} & -1 \\ 0 & 0 & \pi \end{bmatrix}$$

Compute $C = \sin(A)$

$$(a) \det(\lambda I - A) = \det \begin{pmatrix} \lambda+1 & -1 \\ 1 & \lambda+1 \end{pmatrix} \cdot \det \begin{pmatrix} \lambda-2 & 3 \\ 0 & \lambda+2 \end{pmatrix} = (\lambda^2 + 2\lambda + 2)(\lambda^2 - 4)$$

$$= \lambda^4 + 2\lambda^3 - 2\lambda^2 - 8\lambda - 8$$

by Cayley-Hamilton theorem

$$A^4 + 2A^3 - 2A^2 - 8A - 8I = 0$$

$$\Rightarrow B + A - 2I = 0$$

(10 points)

$$\Rightarrow B = 2I - A = \begin{bmatrix} 3 & -1 & 1 & -8 \\ 1 & 3 & -9 & -1 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

If direct multiplication is used then -5 for every mistake.

$$(b) \det(\lambda I - A) = \lambda(\lambda - \pi/2)(\lambda - \pi) \rightarrow \lambda_1 = 0; \lambda_2 = \pi/2; \lambda_3 = \pi$$

$$C = \sin(A) = a_0 I + a_1 A + a_2 A^2 \quad (\text{by Cayley-Hamilton theorem})$$

$$\sin(\lambda_i) = a_0 + a_1 \lambda_i + a_2 \lambda_i^2 \quad i=1, 2, 3 \quad \text{Note: } \sin(0) = 0 \quad \sin(\pi) = 0$$

$$\sin(\pi/2) = 1$$

$$\sin(0) = a_0 \Rightarrow a_0 = 0$$

$$\begin{cases} 1 = a_1 \pi/2 + a_2 \pi^2/4 \\ 0 = a_1 \pi + a_2 \pi^2 \end{cases} \Rightarrow$$

$$a_1 = 4/\pi \quad a_2 = -4/\pi^2$$

(10 points)

$$C = \frac{4}{\pi} \begin{bmatrix} 0 & 1 & 0 \\ 0 & \pi/2 & -1 \\ 0 & 0 & \pi \end{bmatrix} - \frac{4}{\pi^2} \begin{bmatrix} 0 & 1 & 0 \\ 0 & \pi/2 & -1 \\ 0 & 0 & \pi \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & \pi/2 & -1 \\ 0 & 0 & \pi \end{bmatrix}$$

$$\begin{bmatrix} 0 & \pi/2 & -1 \\ 0 & \pi^2/4 & -3\pi/2 \\ 0 & 0 & \pi^2 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 4/\pi & 0 \\ 0 & 2 & -4/\pi \\ 0 & 0 & 4 \end{bmatrix} - \begin{bmatrix} 0 & 2/\pi & -4/\pi^2 \\ 0 & 1 & -6/\pi \\ 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 2/\pi & 4/\pi^2 \\ 0 & 1 & 2/\pi \\ 0 & 0 & 0 \end{bmatrix}$$

Alternative solution of Q26 using diagonalization

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & \pi/2 & -1 \\ 0 & 0 & \pi \end{bmatrix} \quad \det(A - \lambda I) = (0 - \lambda)(\frac{\pi}{2} - \lambda)(\pi - \lambda) = 0$$

$$\Rightarrow \lambda = 0, \frac{\pi}{2}, \pi$$

$$\lambda_1 = 0 \Rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & \pi/2 & -1 \\ 0 & 0 & \pi \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \underline{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda_2 = \frac{\pi}{2} \Rightarrow \begin{bmatrix} -\frac{\pi}{2} & 1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & \frac{\pi}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} x_3 = 0 & x_1 = \alpha \\ -\frac{\pi}{2} x_1 = -x_2 & \\ \frac{\pi}{2} \alpha = x_2 & \end{matrix} \quad \underline{e}_2 = \begin{bmatrix} 1 \\ \pi/2 \\ 0 \end{bmatrix}$$

$$\lambda_3 = \pi \Rightarrow \begin{bmatrix} -\pi & 1 & 0 \\ 0 & -\frac{\pi}{2} & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{matrix} x_3 = \alpha \\ -\frac{\pi}{2} x_2 = \alpha \\ x_1 = -\frac{2}{\pi} \alpha \end{matrix} \quad \begin{matrix} -\pi x_1 = -x_2 \\ x_1 = \frac{x_2}{\pi} = \frac{-2}{\pi^2} \alpha \end{matrix} \quad \underline{e}_3 = \begin{bmatrix} -2/\pi^2 \\ -2/\pi \\ 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & 1 & -2/\pi^2 \\ 0 & \pi/2 & -2/\pi \\ 0 & 0 & 1 \end{bmatrix} \quad Q^{-1} = \frac{1}{\frac{\pi}{2}} \begin{bmatrix} \frac{\pi}{2} & 0 & 0 \\ -1 & 1 & 0 \\ -\frac{1}{\pi} & \frac{2}{\pi} & \frac{\pi}{2} \end{bmatrix}^T = \begin{bmatrix} 1 & -2/\pi & -2/\pi^2 \\ 0 & 2/\pi & 4/\pi^2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Q^{-1} A Q = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \pi/2 & 0 \\ 0 & 0 & \pi \end{bmatrix} = \Lambda \quad \sin A = Q \sin(\Lambda) Q^{-1} \quad \sin(\Lambda) = \begin{bmatrix} \sin 0 & 0 & 0 \\ 0 & \sin \frac{\pi}{2} & 0 \\ 0 & 0 & \sin \pi \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & -2/\pi^2 \\ 0 & \pi/2 & -2/\pi \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2/\pi & 4/\pi^2 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 2/\pi & 4/\pi^2 \\ 0 & 1 & 2/\pi \\ 0 & 0 & 0 \end{bmatrix}$$

Q3. (20 points)

Consider the 4×3 matrix $A = \begin{bmatrix} 2 & 1 & 5 \\ 1 & 0 & 3 \\ 0 & -1 & 1 \\ 2 & 3 & 3 \end{bmatrix}$

- Find a basis for $\text{Row}(A)$
- Find a basis for $\text{Col}(A)$
- Find a basis for $\text{Null}(A)$
- Prove that any vector in $\text{Null}(A)$ is orthogonal to any vector in $\text{Row}(A)$

Note: To have uniformity w.r.t. notation define row vectors such that $A = \begin{bmatrix} r_1^T \\ r_2^T \\ r_3^T \\ r_4^T \end{bmatrix}$ where

$$r_1 = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}, r_2 = \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, r_3 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, r_4 = \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}.$$

(a) $\text{Row}(A) = \text{Span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix} \right\}$ note: $\begin{cases} r_4 = r_1 - 2r_3 \\ r_1 = 2r_2 - r_3 \end{cases}$ $r_2^T r_3 \neq 0$

5 points $= \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\}$ $\{r_2, r_3\}$ is a basis for $\text{Row}(A)$

(b) $\text{Col}(A) = \text{Span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \\ 1 \\ 3 \end{bmatrix} \right\}$ $3v_1 - v_2 = v_3$

5 points $v_1 \quad v_2 \quad v_3$ $\{v_1, v_2\}$ is a basis for $\text{Col}(A)$ $v_1^T v_2 \neq 0$

(c) by (b) $3v_1 - v_2 - v_3 = \underline{0}$ i.e. $A \cdot \begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix} = \underline{0}$

a basis for $\text{Null}(A) = \left\{ \begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix} \right\}$ i.e. $\text{Null}(A) = \text{Span} \left\{ \begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix} \right\}$

5 points

(d) We need to prove that $\alpha \begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix}$ is orthogonal to $\beta_1 \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \beta_2 \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$

for any α, β_1, β_2 constants

$$\alpha \begin{bmatrix} 3 & -1 & -1 \end{bmatrix} \left(\beta_1 \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \beta_2 \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right) = 0 \quad \checkmark$$

5 points

Alternative solutions

$$a) \begin{bmatrix} 2 & 1 & 5 \\ 1 & 0 & 3 \\ 0 & -1 & 1 \\ 2 & 3 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 5 \\ 0 & -1 & 1 \\ 2 & 3 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \\ 0 & 3 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Row}(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$$

In fact any two rows of A are OK because $r=2$.

b) Since $r=2$ any two columns are OK.
The pivot columns may be used.

$$\text{Col}(A) = \text{Span} \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \\ 3 \end{bmatrix} \right\}$$

c) Using the ref form in a)

$$x_3 = \alpha$$

$$x_2 - x_3 = 0 \Rightarrow x_2 = \alpha$$

$$x_1 + 3x_3 = 0 \Rightarrow x_1 = -3\alpha$$

$$\Rightarrow \text{Null}(A) = \text{Span} \left\{ \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \right\}$$

d) Same

Q4. (20 points)

- Prove that the determinant of a square matrix whose columns are orthonormal is equal to +1 or -1.
- Find the projection of the vector $\begin{bmatrix} 1 \\ 5 \end{bmatrix}$ onto the line defined by $x_1 + x_2 = 0$
- Prove that the set of 2×2 matrices whose determinants are 0 does not form a vector space.
- Let A be an $n \times n$ matrix such that $Ae_1, Ae_2, \dots, Ae_n$ are linearly independent where the vectors $e_1, e_2, \dots, e_n$ are linearly independent. Prove that $\det(A) \neq 0$.
- Let A_1 be an $n \times n$ invertible matrix, and A_3 be an $m \times m$ invertible matrix where $m \neq n$. For any given $n \times m$ matrix A_2 , find the inverse of $\begin{bmatrix} A_1 & A_2 \\ 0 & A_3 \end{bmatrix}$.

(a) $A = [\underline{v}_1 \mid \dots \mid \underline{v}_n]$ such that $\underline{v}_i^T \underline{v}_j = 0 \quad i \neq j$ and $\underline{v}_i^T \underline{v}_i = 1$

So $A^T A = I \quad \text{i.e.} \quad A^{-1} = A^T \quad \det(A^T) = \det(A)$

$\Rightarrow (\det(A))^2 = 1 \quad \Rightarrow (\det A) = \pm 1 \quad 4 \text{ points}$

(b) $\begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \Rightarrow x_2 = -x_1 \rightarrow \text{line} \equiv \text{Span} \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\} = \text{Span} \left\{ \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} \right\}$
 $\underline{y} =$ orthonormal basis for the line

$\underline{x} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$ projection of $\underline{x}$ onto $\underline{y}$ is
 $(1 \ 5) \begin{bmatrix} 1 \\ -1 \end{bmatrix} \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} = -4 \cdot \begin{bmatrix} 1/2 \\ -1/2 \end{bmatrix} = \begin{bmatrix} -2 \\ +2 \end{bmatrix} \quad 4 \text{ points}$

(c) Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \quad \det(A) = 0 \quad \det(B) = 0$
 but $A+B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \det(A+B) \neq 0$
 So the space is not closed under addition 4 points

(d) $Q = [Ae_1 \mid \dots \mid Ae_n]$ has rank n because $\{Ae_1, \dots, Ae_n\}$ are LI $\Rightarrow \det(Q) \neq 0$
 $E = [e_1 \mid \dots \mid e_n]$ has rank n because $\{e_1, \dots, e_n\}$ are LI $\Rightarrow \det(E) \neq 0$
 $Q = A E \quad \det(A) = \frac{\det Q}{\det E} \neq 0 \quad 4 \text{ points}$

(e) $A = \begin{bmatrix} A_1 & A_2 \\ 0 & A_3 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} A_1^{-1} & -A_1^{-1} A_2 A_3^{-1} \\ 0 & A_3^{-1} \end{bmatrix}$
 Check that $A \cdot A^{-1} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} \quad A^{-1} A = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} \quad \checkmark \quad 4 \text{ points}$

Alternative solutions

a) Same

b) Same

c) Same

d) Same

$$e) \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} \begin{bmatrix} A_1 & A_2 \\ 0 & A_3 \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$$

$$B_1 A_1 = I \Rightarrow B_1 = A_1^{-1}$$

$$B_3 A_1 = 0 \Rightarrow B_3 = 0$$

$$B_3 A_2 + B_4 A_3 = I \Rightarrow B_4 A_3 = I \Rightarrow B_4 = A_3^{-1}$$

$$B_1 A_2 + B_2 A_3 = 0 \Rightarrow A_1^{-1} A_2 + B_2 A_3 = 0$$

$$A_1^{-1} A_2 A_3^{-1} + B_2 A_3 A_3^{-1} = 0$$

$$A_1^{-1} A_2 A_3^{-1} + B_2 = 0$$

$$B_2 = -A_1^{-1} A_2 A_3^{-1}$$

$$\text{Inverse} = \begin{bmatrix} A_1^{-1} & -A_1^{-1} A_2 A_3^{-1} \\ 0 & A_3^{-1} \end{bmatrix}$$

Q5. (20 points) Consider an $n \times n$ matrix A with real entries satisfying the following properties

(i) $A^T = A$,

(ii) $x^T A x > 0$ for all $x \in \mathbb{R}^n$ such that $x \neq \underline{0}$,

(iii) eigenvalues $\{\lambda_1, \dots, \lambda_n\}$ of A are distinct.

(a) Prove that the eigenvectors of A are orthogonal.

(b) Prove that $\lambda_i > 0$ for all i .

(c) Prove that A^{-1} exists.

(d) Prove that the eigenvalues of A^{-1} are $\{\lambda_1^{-1}, \dots, \lambda_n^{-1}\}$.

Let $A \underline{e}_i = \lambda_i \underline{e}_i$

eigenvalues $\lambda_1, \dots, \lambda_n$
eigenvectors $\underline{e}_1, \dots, \underline{e}_n$

(a) (5pts) $\underline{e}_j^T A \underline{e}_i = \lambda_i \underline{e}_j^T \underline{e}_i$
 $(A^T \underline{e}_j)^T \underline{e}_i = (A \underline{e}_j)^T \underline{e}_i = \lambda_j \underline{e}_j^T \underline{e}_i = \lambda_i \underline{e}_j^T \underline{e}_i \quad (\lambda_j - \lambda_i) \underline{e}_j^T \underline{e}_i = 0$
 Since $\lambda_i \neq \lambda_j$ for $i \neq j$ we have $\underline{e}_j^T \underline{e}_i = 0$ for $i \neq j$
 $\Rightarrow$ eigenvectors are orthogonal

(b) (5pts) $\underline{e}_i^T A \underline{e}_i > 0 \quad \forall i$ but $\underline{e}_i^T A \underline{e}_i = \underline{e}_i^T (\lambda_i \underline{e}_i) = \lambda_i \underline{e}_i^T \underline{e}_i = \lambda_i \|\underline{e}_i\|^2$
 Since $\underline{e}_i \neq \underline{0}$, $\|\underline{e}_i\|^2 \neq 0$ so $\underline{e}_i^T A \underline{e}_i \frac{1}{\|\underline{e}_i\|^2} = \lambda_i > 0 \quad \forall i$

(c) (5pts) $A [\underline{e}_1 \dots \underline{e}_n] = [\underline{e}_1 \dots \underline{e}_n] \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$
 $\det(A) = \lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n$ since each $\lambda_i > 0$ $\det(A) \neq 0$
 $\Rightarrow A^{-1}$ exists

(d) (5pts) $A \underline{e}_i = \lambda_i \underline{e}_i \quad A^{-1} \text{ exists} \Rightarrow$
 $A^{-1} (A \underline{e}_i) = \lambda_i A^{-1} \underline{e}_i \Rightarrow \underline{e}_i = \lambda_i A^{-1} \underline{e}_i$
 $\Rightarrow \frac{1}{\lambda_i} \underline{e}_i = A^{-1} \underline{e}_i$ so $(\frac{1}{\lambda_i}, \underline{e}_i)$ is an eigenpair for A^{-1}
 $\Rightarrow$ eigenvalues of $A^{-1} = \{\lambda_1^{-1}, \dots, \lambda_n^{-1}\}$.